Prof. V. B. Shah Institute of Management & R. V. Patel College of Commerce (Eng. Med.) & V. L. Shah College of Commerce (Guj. Med.) & Sutex Bank College of Computer Applications Science, Amroli

VEER NARMAD SOUTH GUJARAT UNIVERSITY, SURAT

PROJECT REPORT

ON

Techify - E-commerce Platform

SUBMITTED BY:

Jenish Bhlala S.

Pruthviraj Sakariya S.

Raj patoliya R.

Roll No: 412 , 433 , 470

GUIDED BY:

Asst. prof. Ashish Solanki

Exam No: 6053,6359,6311

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1.Introduction

1.1 Project Description :-

- Techify is a modern, responsive full-stack e-commerce platform designed to provide a seamless online shopping experience for technology products and electronics.
- It serves everyday consumers and tech enthusiasts seeking a reliable, secure, and user-friendly storefront to discover, save, and purchase products.
- The platform emphasizes an intuitive user interface, efficient product management, secure cart and checkout processes, and responsive design.
- Techify consists of two primary access levels:
    I. Admin Panel
    II. User Panel

❖Admin Panel :-

- Requires login with an administrator account.
- Admin can add, view, update, and delete products from the catalog.
- Admin can view, manage, and delete registered user accounts.
- Admin can view user orders and update their order status (Processing, Shipped, Delivered).

- Admin can read and manage customer inquiries and messages submitted via the Contact Us form.

❖User Panel :-

- Requires login with a user account (guests may be able to browse).
- Users can view and update their profile, including their shipping address.
- Users can browse, search, and filter products by category and brand.
- Users can add products to an active shopping cart or save them to a wishlist for later.
- Users can proceed to checkout, place orders securely, and view their past order history.
- Users can send inquiries, complaints, or suggestions to the admin using the Contact Us form.

1.2 Project Profile :-

Techify	
Project Title	Techify
Fontend	HTML , CSS, Anguler, Tailwind css
Backend	Node Js, Express Js
Database	MongoDB
Platform	Vs Code, Mongo DB Compass, Stitch, Git Hub
Documantaion Tools	MS Word, Figma
Guide Name	Mr. Ashish Solanki

2. Objectives

❖Strategic Objectives:

- Build a credible and performant foundation for a modern e-commerce platform encompassing customers and administrators.
- Position the platform for seamless future payment integrations (Stripe, Razorpay, etc.) and promotional features (coupons, flash sales).
- Enable data-driven business decisions through tracking of order volumes, inventory levels, and user registration trends.

❖Commerce & Sales Objectives:

- Deliver a clear and engaging product catalog with rich details, variant capabilities, and high-quality image support.
- Encourage conversions via an intuitive cart management system and seamless multi-step checkout process.
- Support dynamic pricing models including calculated discount percentages and virtual final prices.

❖Customer Experience Objectives:

- Fast, intuitive product discovery with filtering and robust search capabilities.
- Provide users with continuous engagement through Wishlist functioning and persistent Cart states.

- Ensure mobile-friendly layout capabilities (responsive component patterns) for on-the-go shopping.

❖Operational / Admin Objectives:

- Centralized management of the product catalog (adding, updating pricing, and managing stock levels).
- Streamlined order fulfillment tracking and status manipulation.
- Effective customer support handling through centralized Contact Message management (tracking read/unread/replied statuses).
- User and role management to oversee customer bases and assign administrative privileges.

❖Technical Objectives:

- Strict separation of concerns (Angular SPA frontend + Node/Express API backend) for independent maintainability and scaling.
- Robust relational data modeling using MongoDB/Mongoose (handling virtual fields, references, and embedded documents).
- Secure authentication and session handling utilizing industry standards (bcrypt password hashing, JWT tokens).
- Modular routing and service-oriented frontend architecture to simplify future feature implementations.

❖Data & Analytics Objectives:

- Capture core metrics (total revenue, total orders, low inventory alerts) for internal dashboards.
- Store historical transaction snapshots (price and product details at the exact time of purchase) to ensure data integrity over time.
- Prepare schema for future personalization (behavioral tracking, viewed products, personalized recommendations).

❖Growth & Retention Objectives:

- Foster repeat purchases by retaining user shipping profiles, wishlists, and seamless re-ordering paths.
- Support trust signals by implementing robust product descriptions, categorizations, and a secure checkout flow.
- Optimize product discovery through structured metadata (categories, brands, custom feature maps).

❖Scalability & Future Roadmap Objectives:

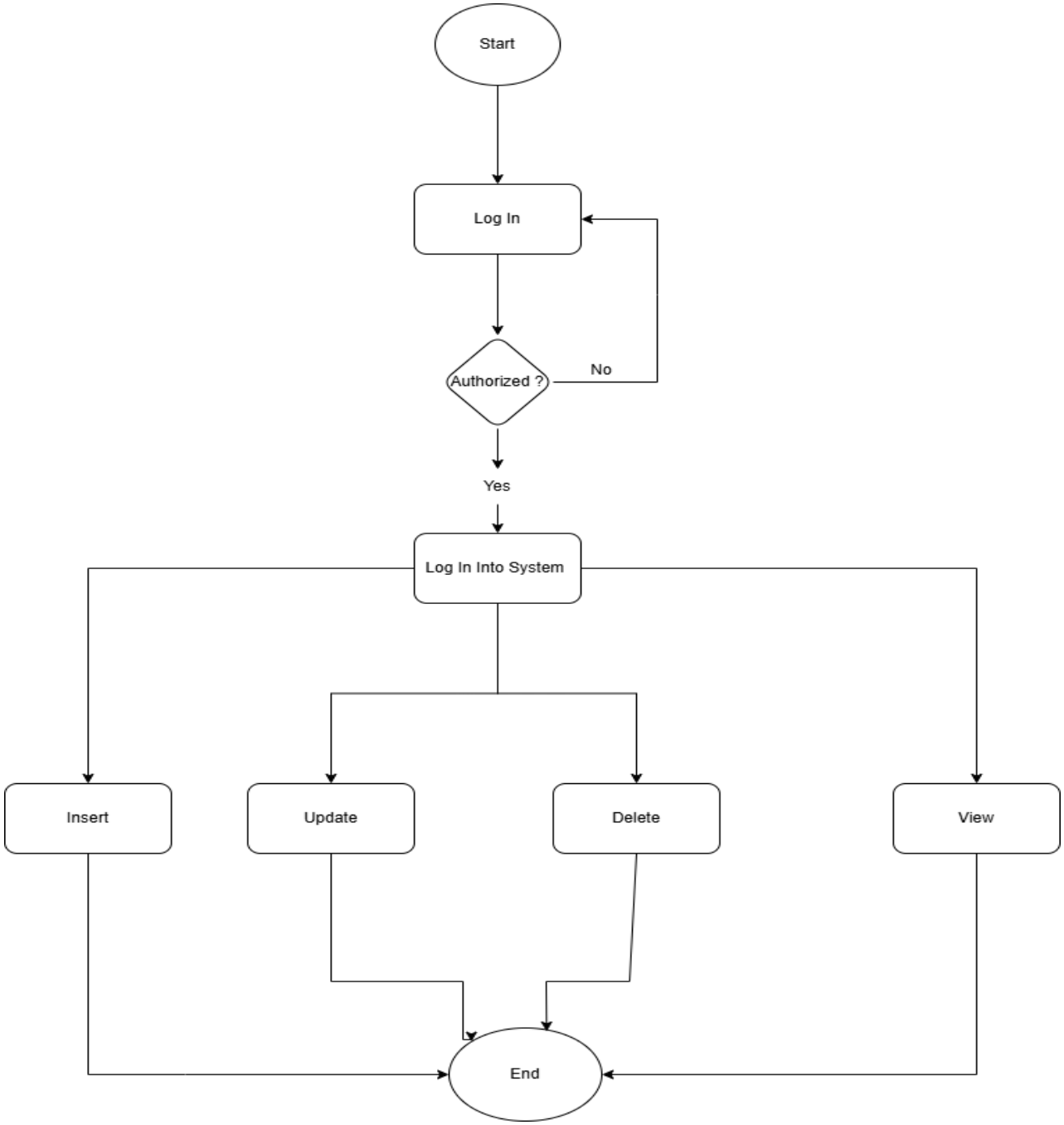
- Plug-in ready architecture for: user product reviews, rating systems, multi-vendor support, and newsletter subscriptions.
- Abstract media handling to allow future integration with cloud image delivery networks (e.g., Cloudinary, AWS S3).
- Implement advanced stock reservation systems to prevent overselling during high-traffic flash sales.

❖Compliance & Security Objectives:

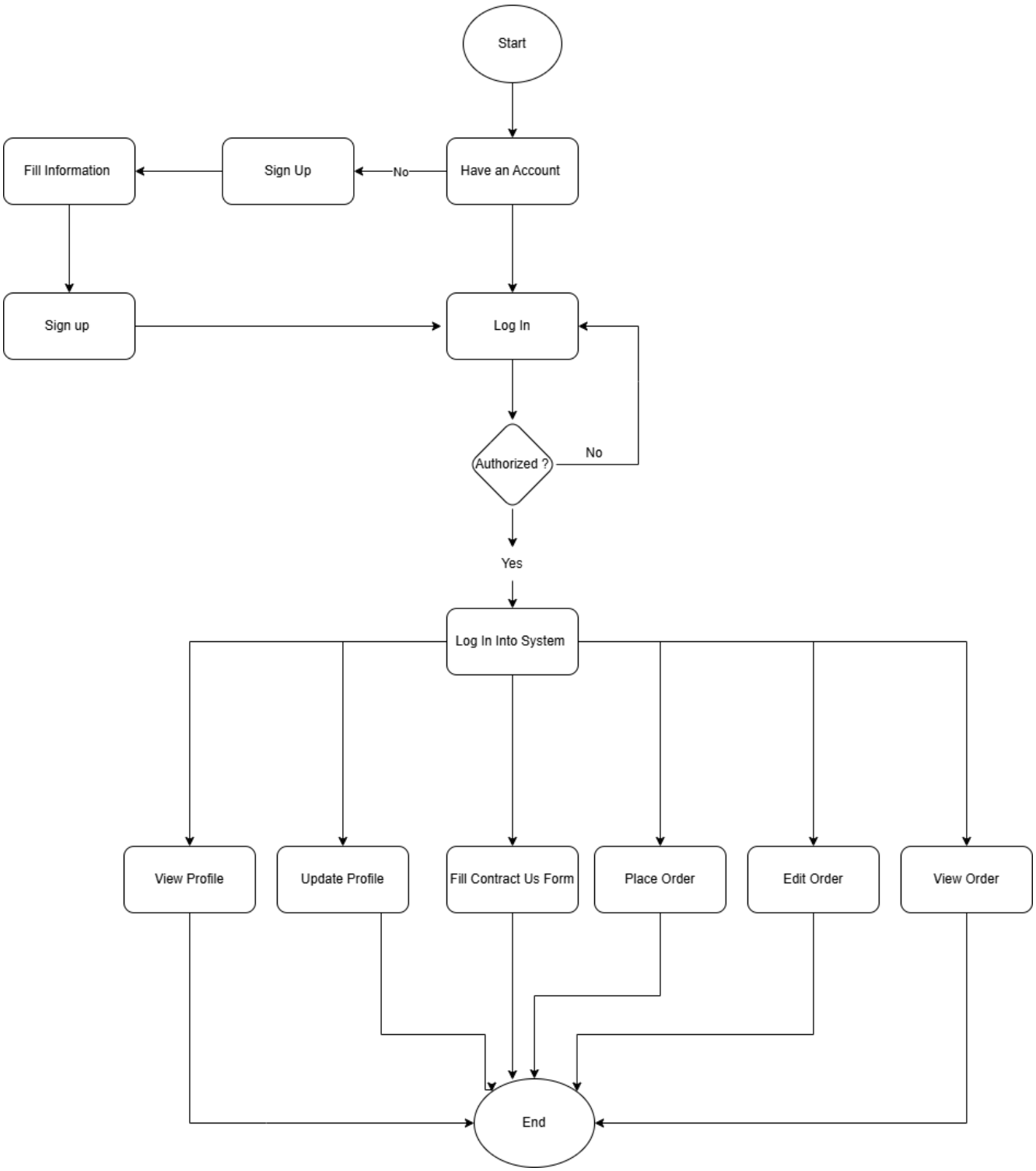
- Enforce strict Route Guards and JWT-based access boundaries (Admin vs User operations).
- Ensure sensitive customer data (passwords, addresses) is correctly hashed and transmitted securely.
- Mitigate standard vulnerabilities (e.g., injection, cross-site scripting) across product entry and user input forms.

3. System Flow Diagram

❖Admin Process Flow Diagram:-

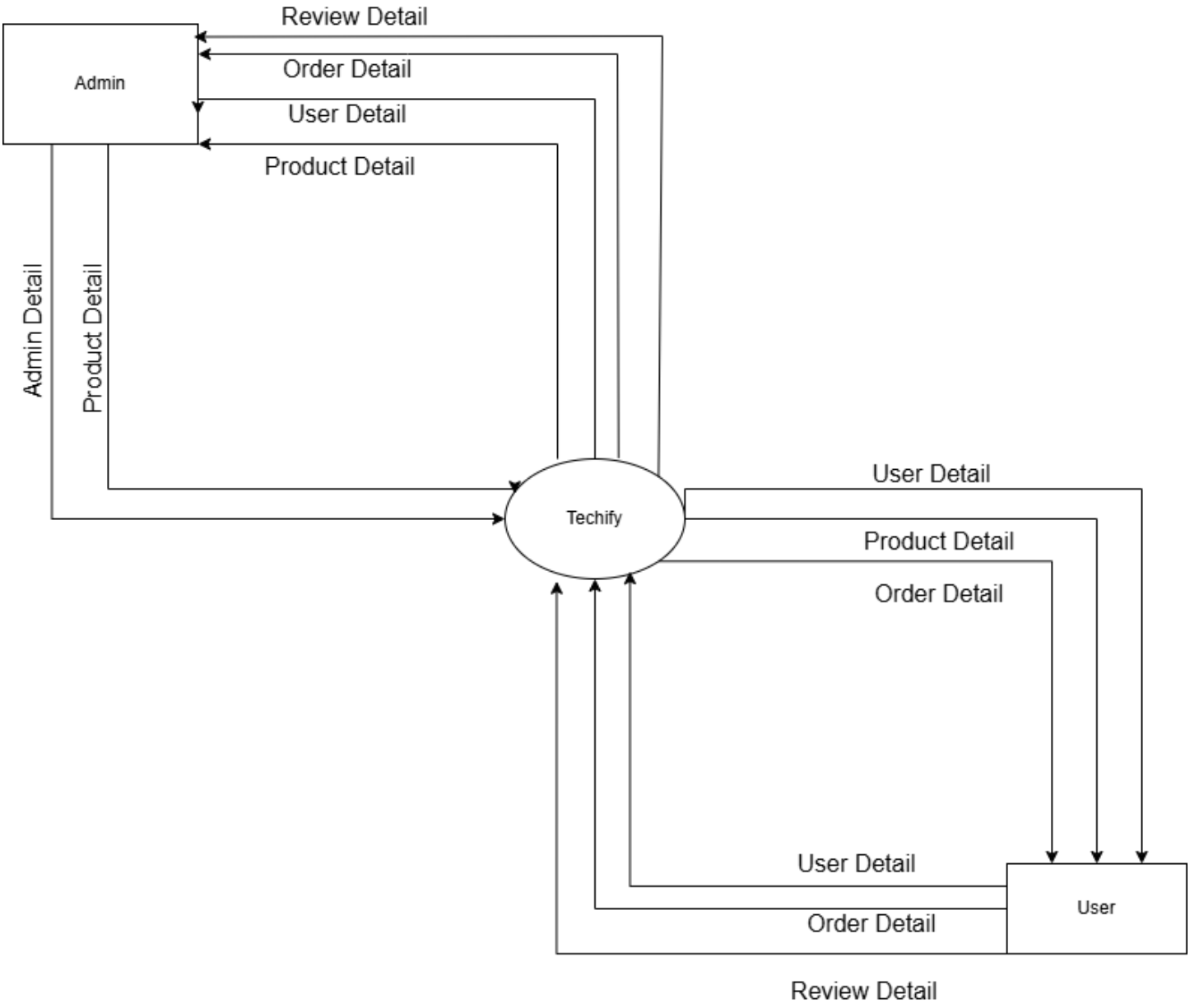


❖User Process Flow Diagram:-

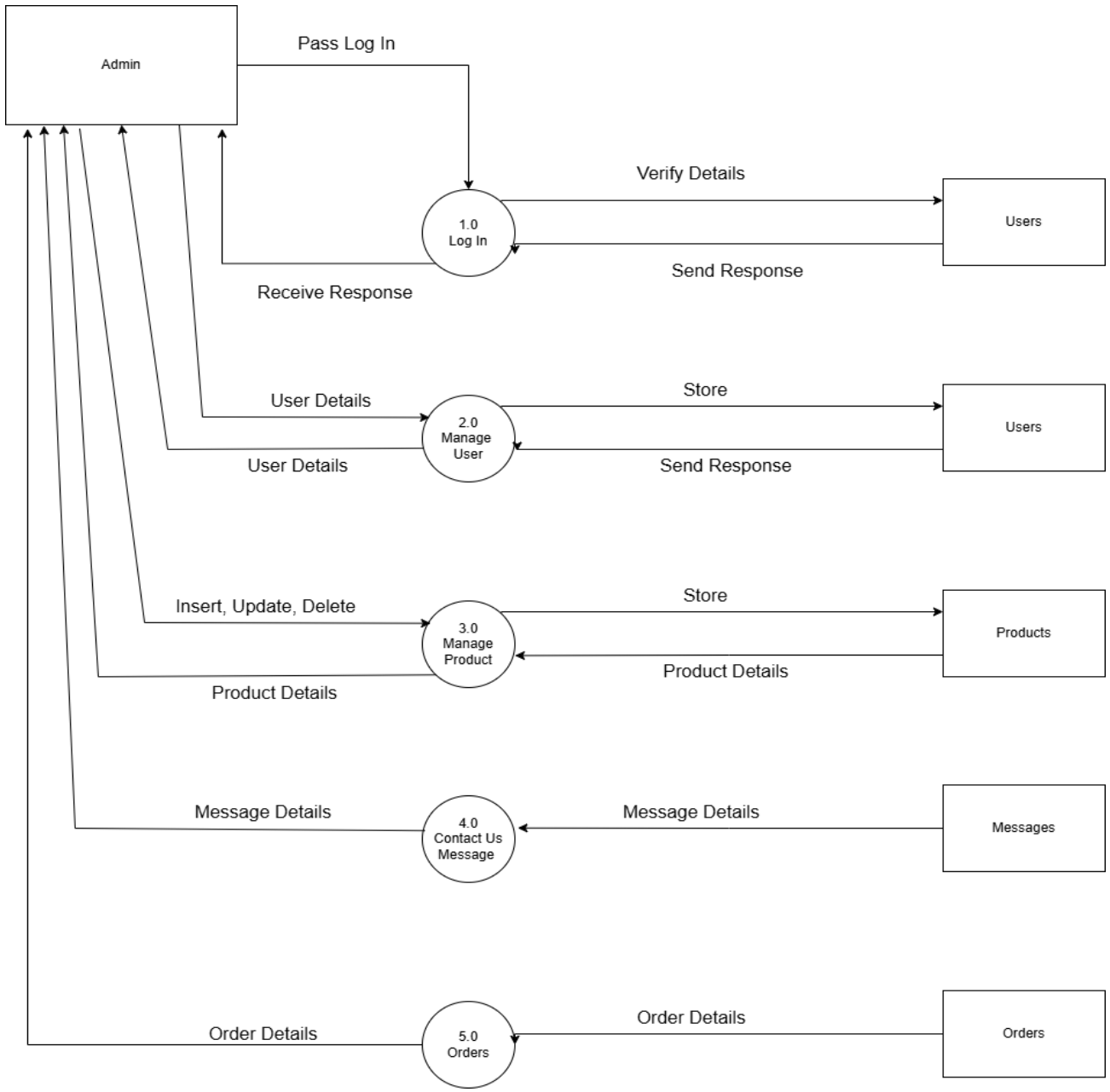


4. DFD / UML Diagram

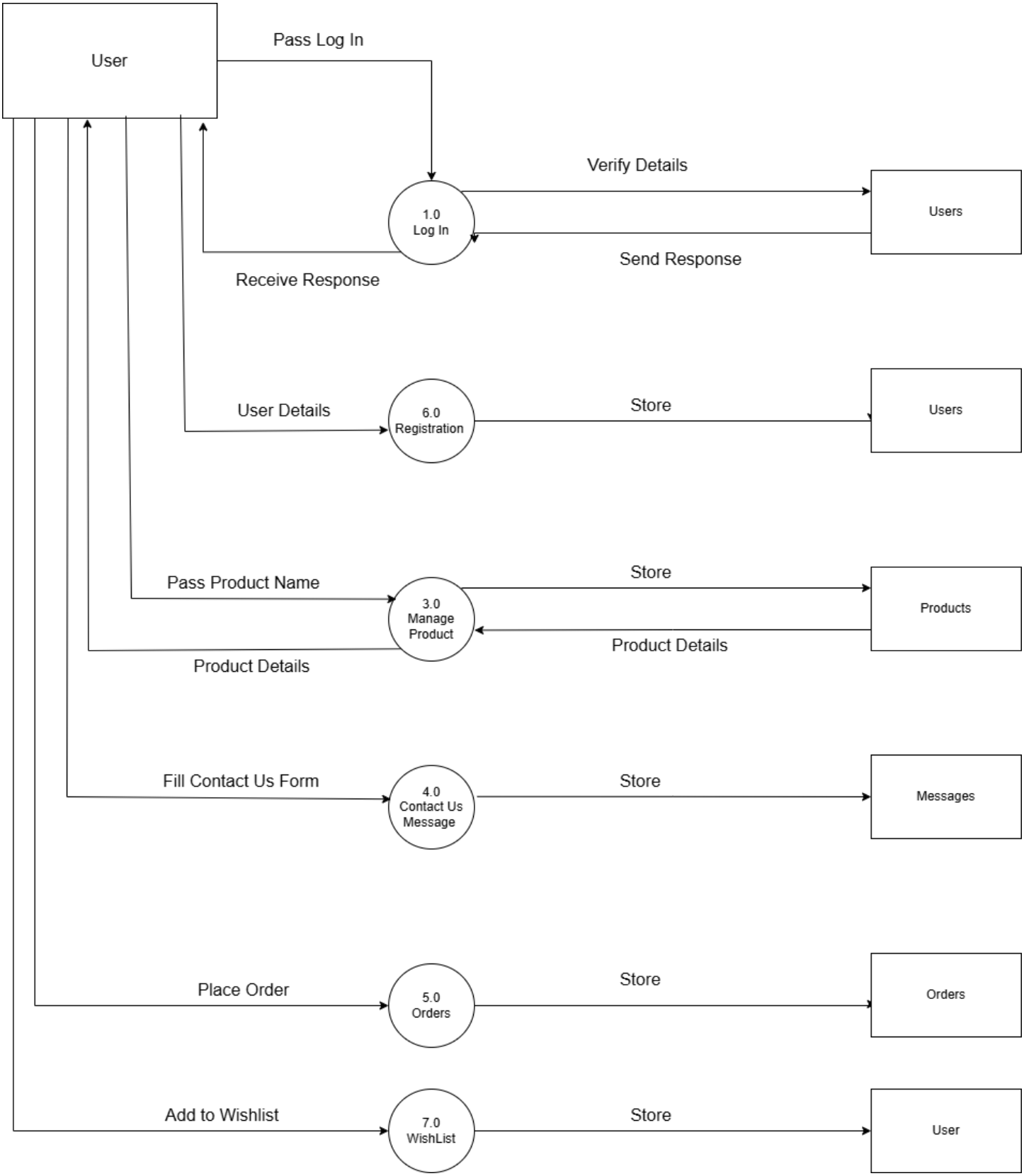
❖0 Level Diagram(Context Diagram) :



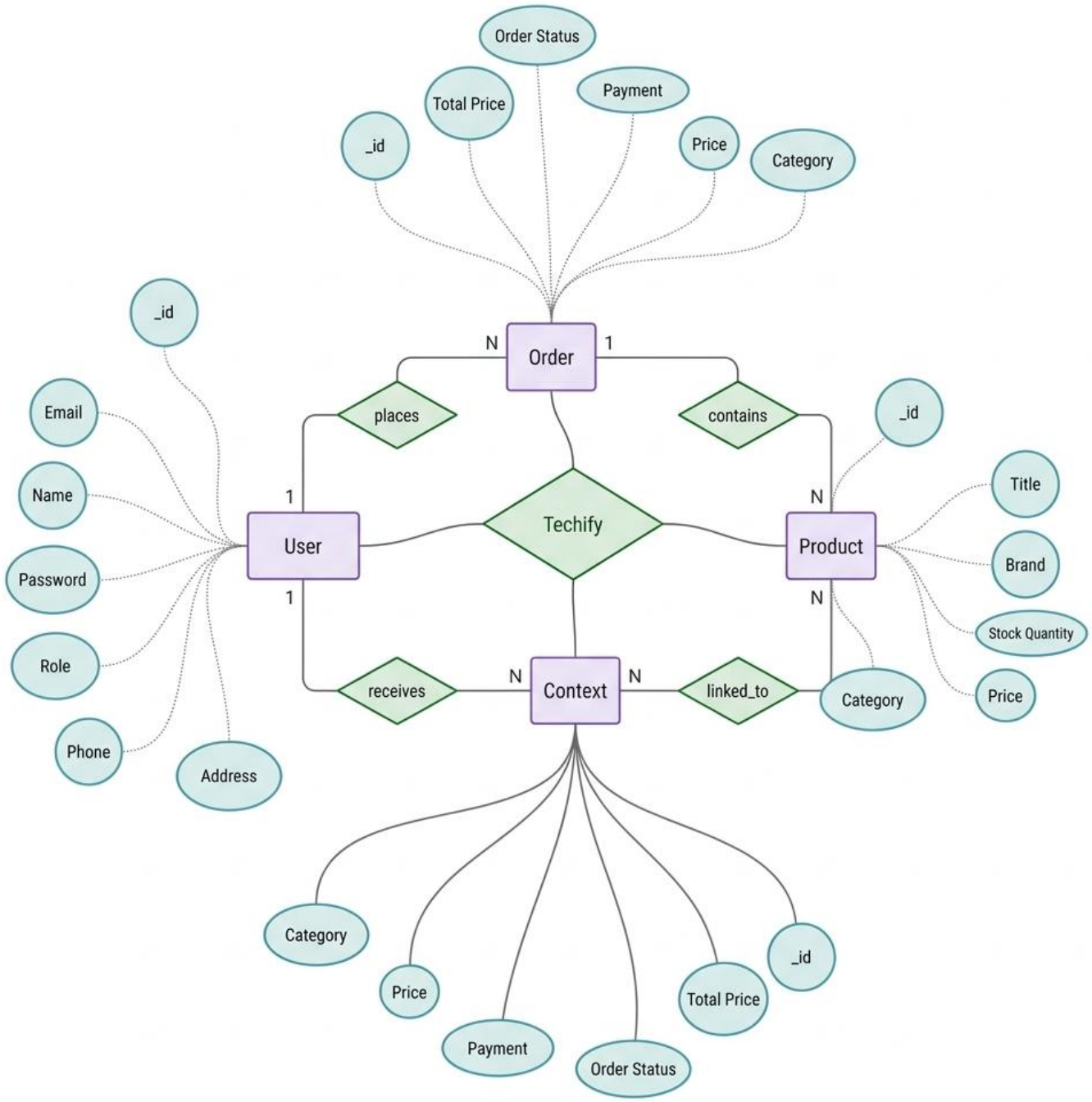
❖1st Level Admin Diagram :-



❖1st Level User Diagram : -



5. ER Diagram


Order Status
Total Price
Payment
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Price
Category
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Email
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Techify
N
Title
User
Product
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Brand
1
N
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Role
receives
N
Context
N
linked_to
Category
Price
Phone
Address
Category
Price
Payment
Order Status
Total Price
_id

6. Database Design

❖User Collection:-

Field	Type	Description
name	String	User full name
email	String	Unique login email
password	String	Encrypted password
role	Enum	user/admin access role
cart	Array	Embedded items with productId and quantity
wishlist	Array<ObjectId>	Saved product references

❖Product Collection :-

Field	Type	Description
title	String	Product title
description	String	Detailed product information
category	String	Product category
brand	String	Brand name
price	Number	Base price
discountPercentage	Number	Discount value in %
finalPrice	Virtual	Computed discounted price
stockQuantity	Number	Current stock

❖Order Collection :-

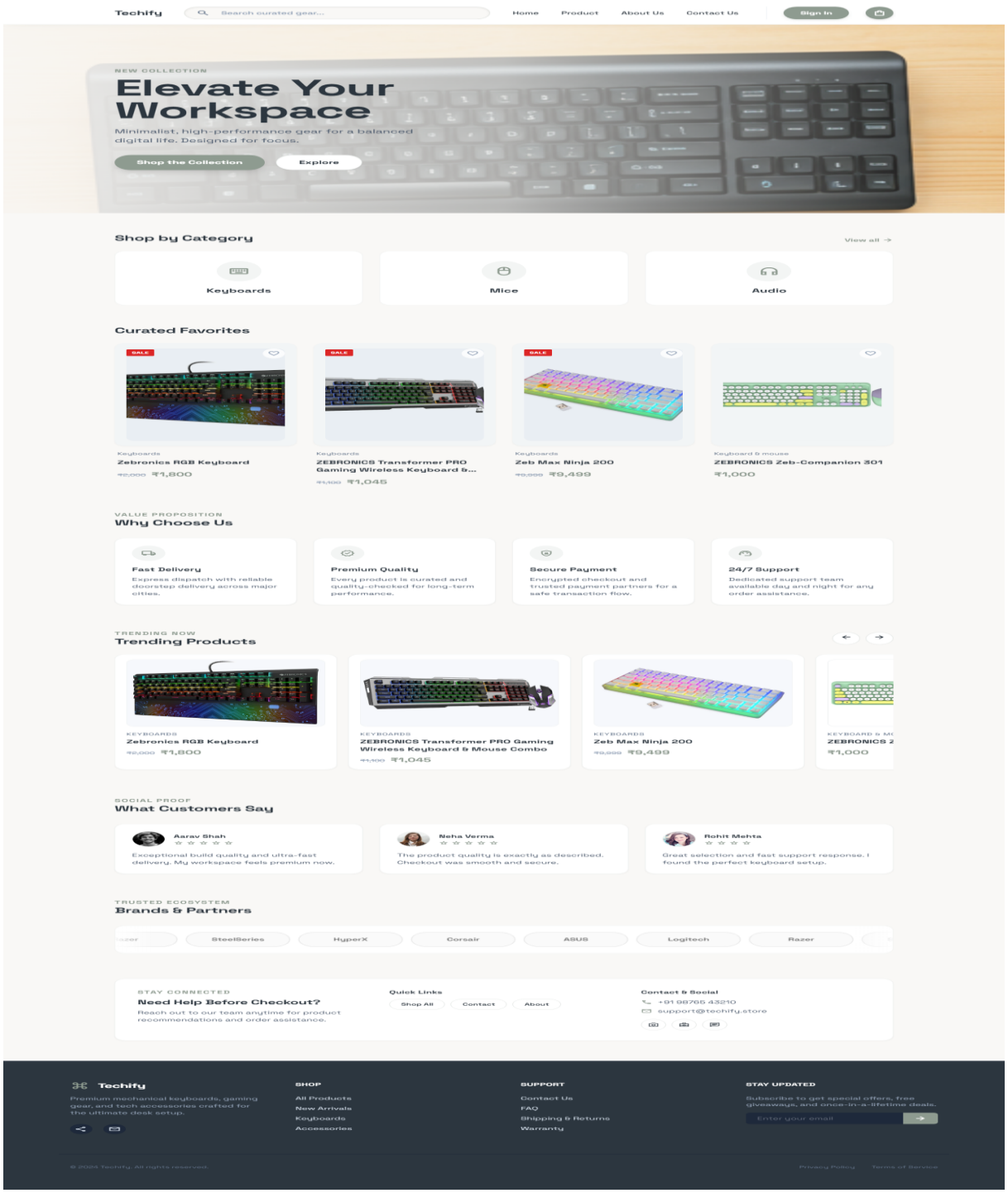
Field	Type	Description
userId	ObjectId	Reference to User
orderItems	Array	Snapshot of purchased products
shippingAddress	Object	Delivery details
paymentMethod	String	Selected payment mode
totalPrice	Number	Final payable amount
orderStatus	Enum	Processing/Shipped/Delivered

❖Message Collection :-

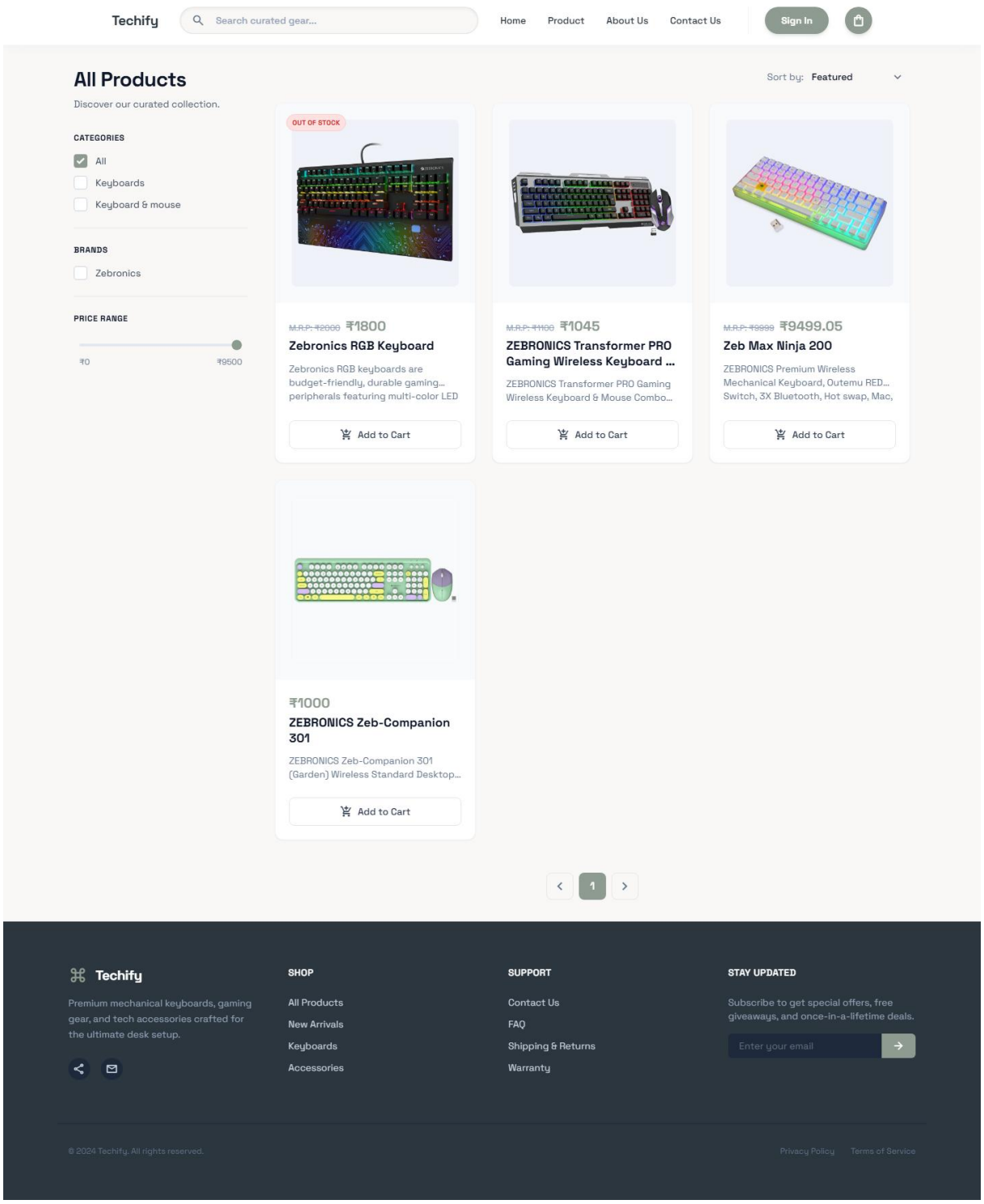
Field	Type	Description
name	String	Sender name
email	String	Sender email
subject	String	Query subject
message	String	Query body
status	Enum	unread/read/replied

7. System Design

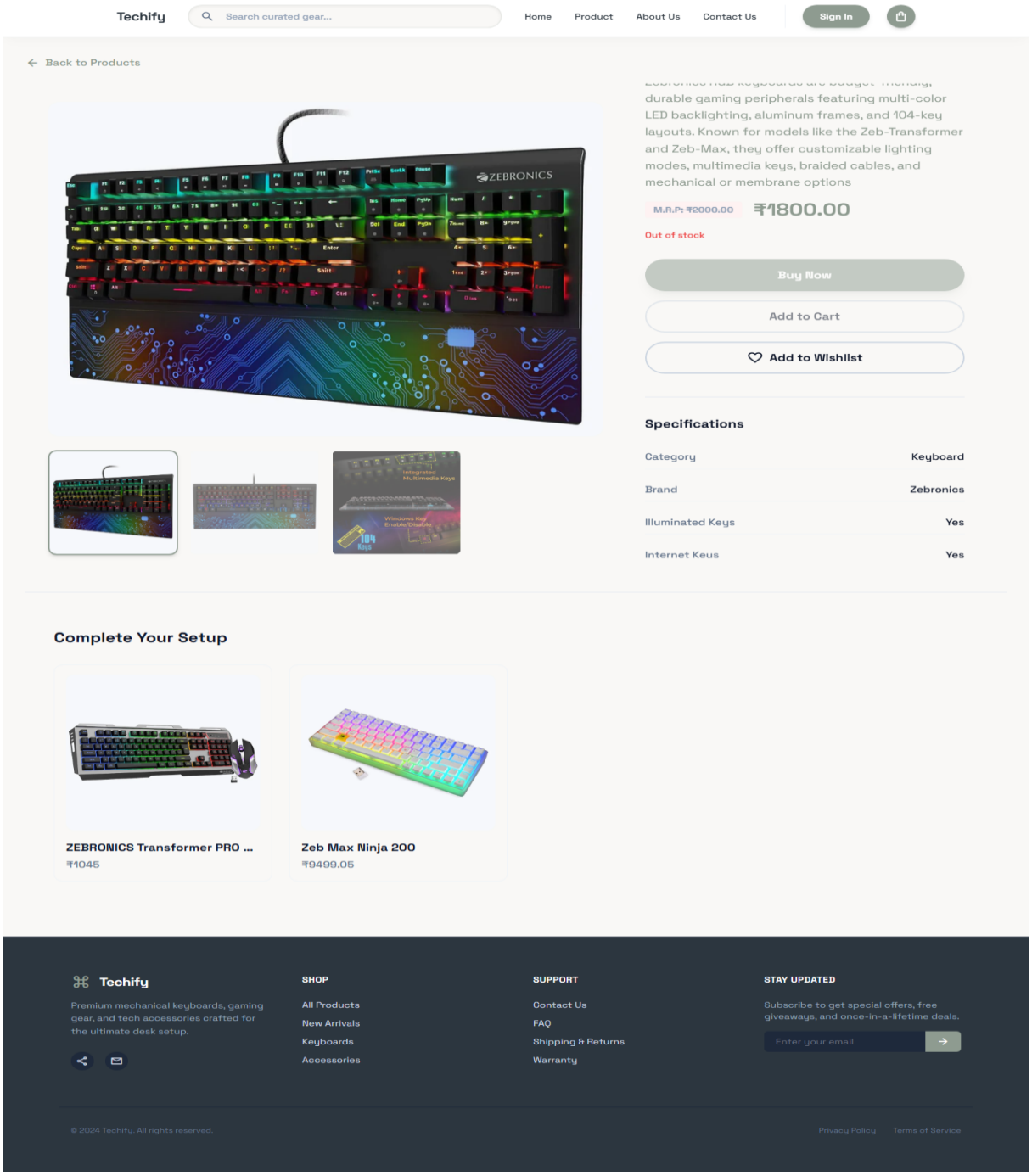
❖Home Page :-



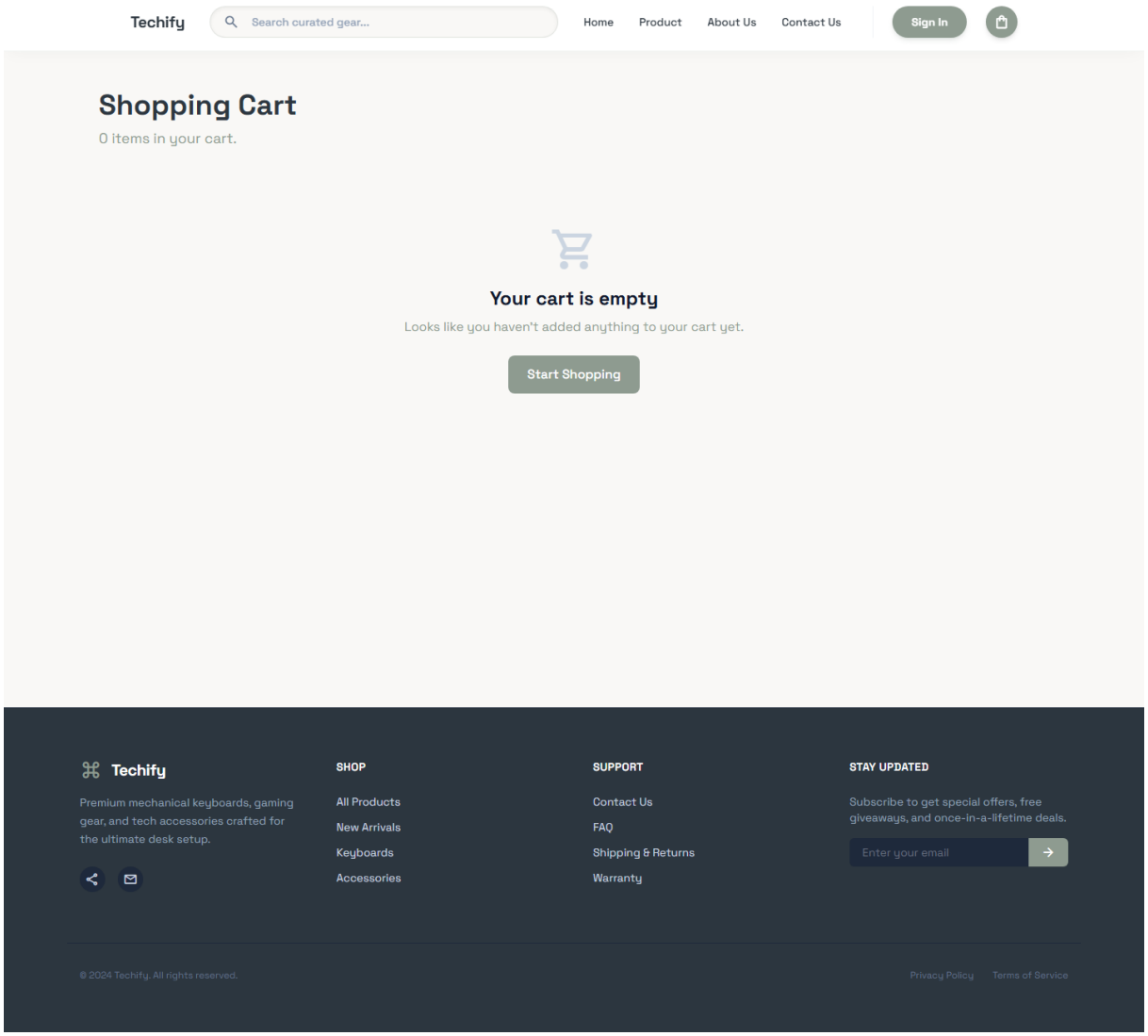
❖Products Page:-



❖Product Detail Page :-



❖Cart Page :-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About Us
Contact Us
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Your cart is empty
Looks like you haven't added anything to your cart yet.
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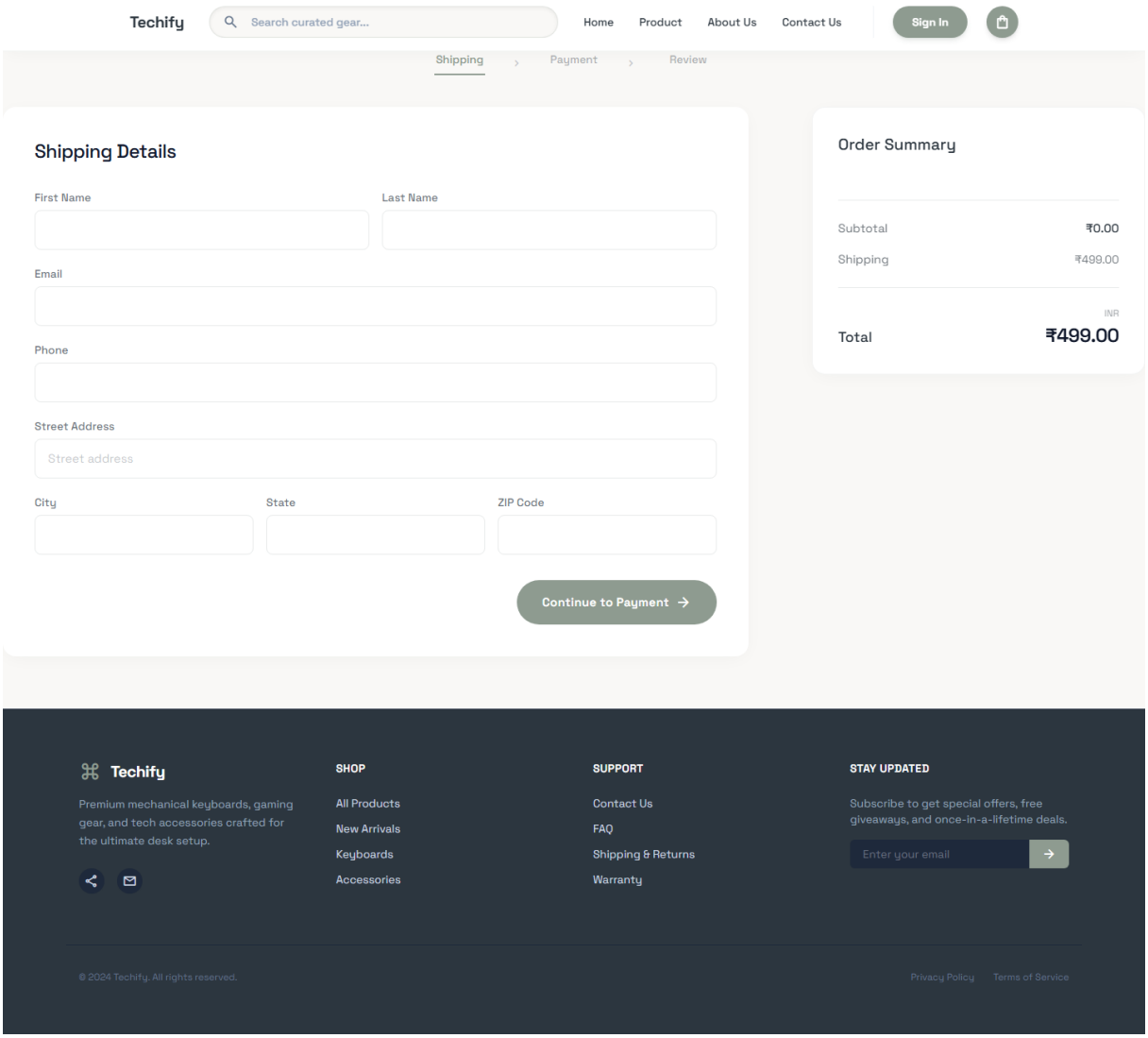
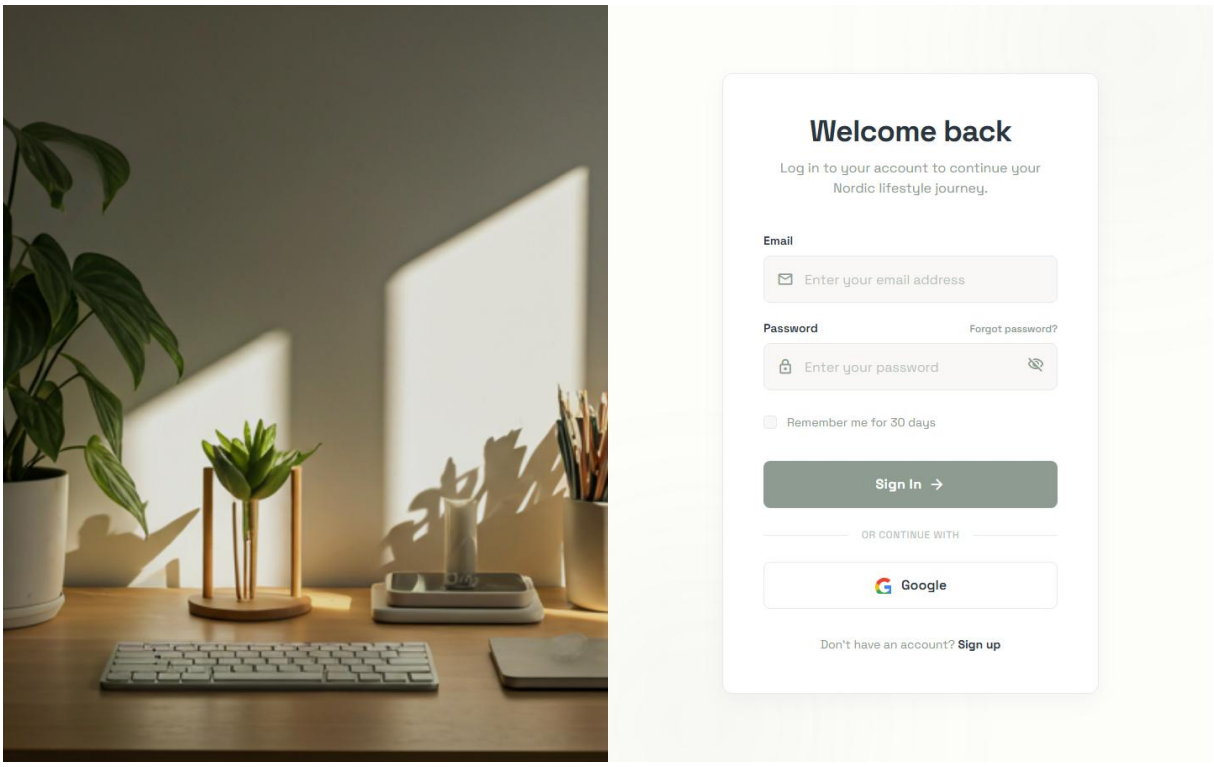
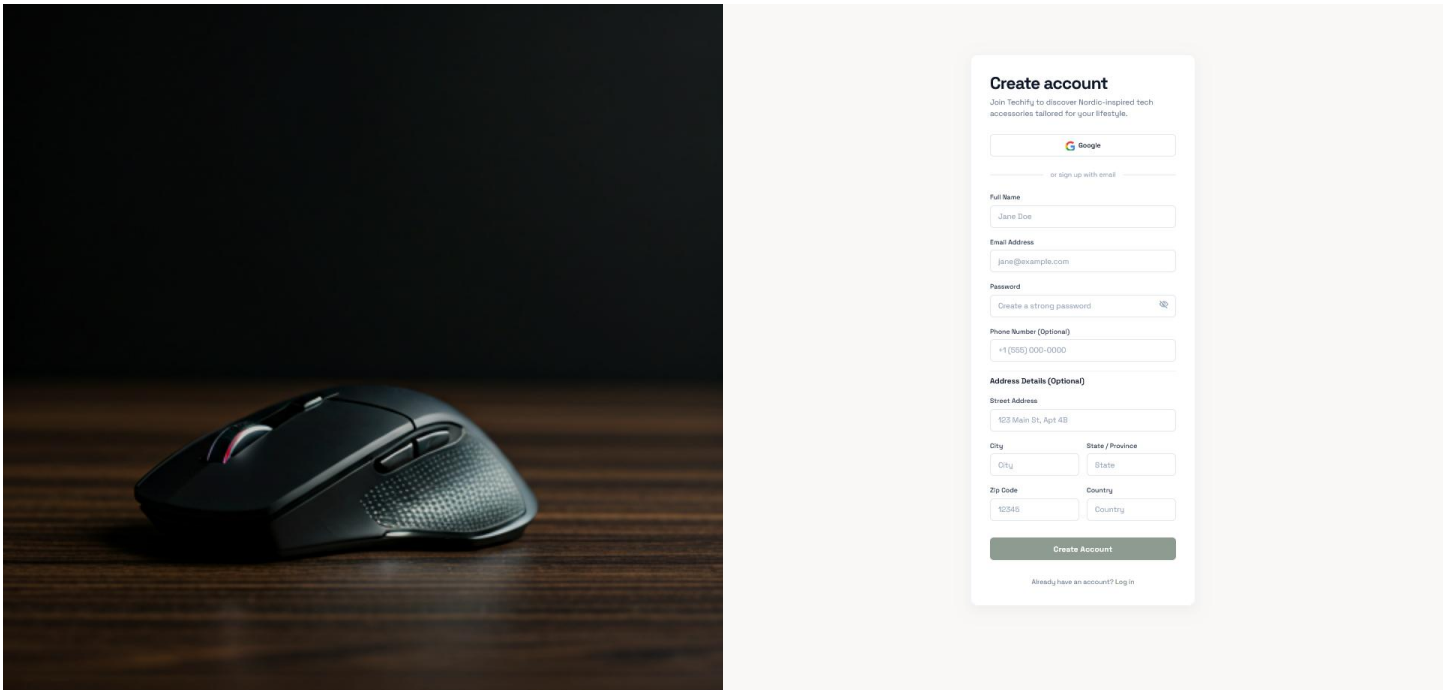
❖Checkout Page :-



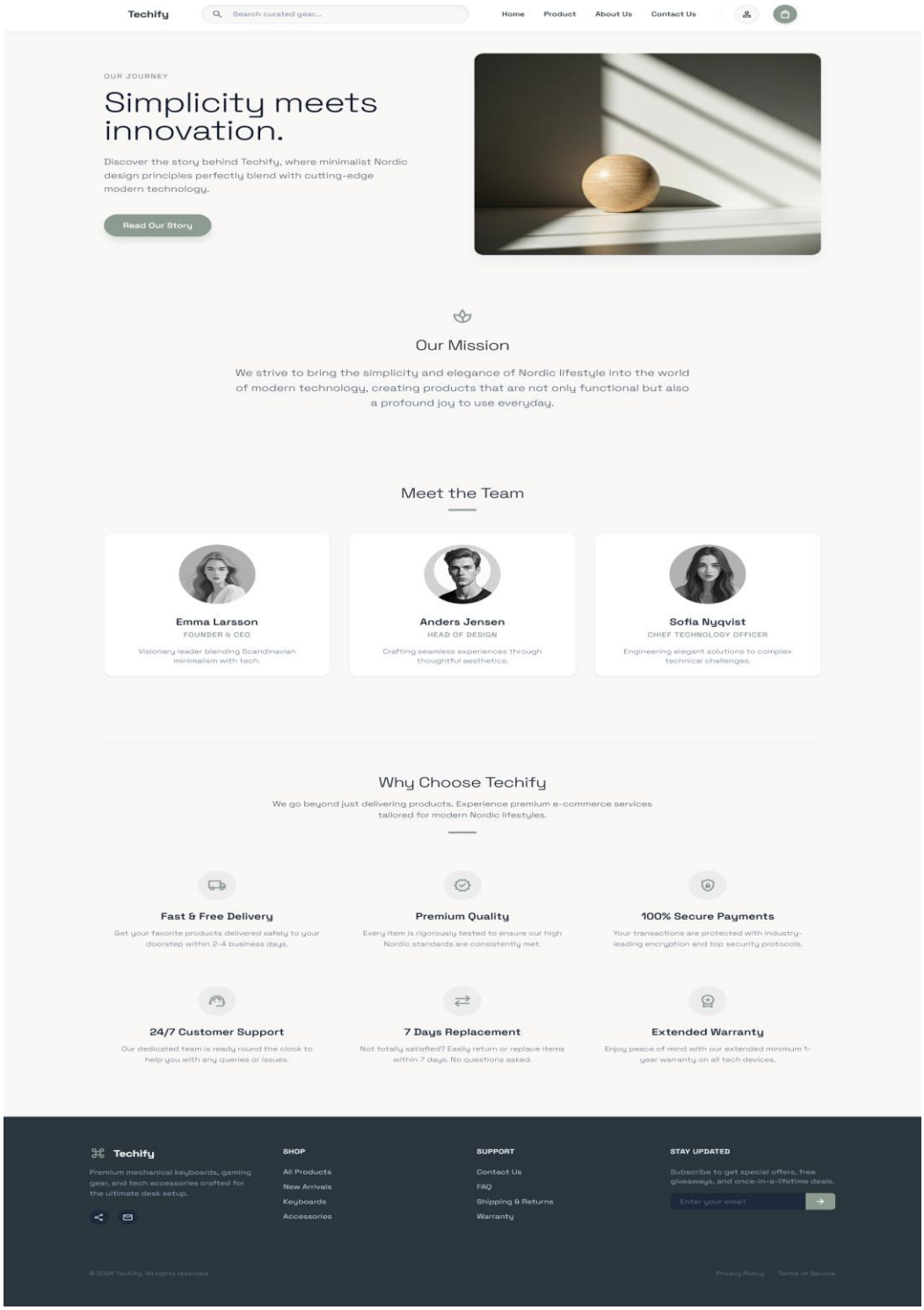
❖ Login Page :-



❖ Register Page:-



❖About Page:-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Contact Us
OUR JOURNEY
Simplicity meets innovation.
Discover the story behind Techify, where minimalist Nordic design principles perfectly blend with cutting-edge modern technology.
Read Our Story
Our Mission
We strive to bring the simplicity and elegance of Nordic lifestyle into the world of modern technology, creating products that are not only functional but also a profound joy to use everyday.
Meet the Team
Emma Larsson
FOUNDER & CEO
Visionary leader blending Scandinavian minimalism with tech.
Anders Jensen
HEAD OF DESIGN
Crafting seamless experiences through thoughtful aesthetics.
Sofia Nyqvist
CHIEF TECHNOLOGY OFFICER
Engineering elegant solutions to complex technical challenges.
Why Choose Techify
We go beyond just delivering products. Experience premium e-commerce services tailored for modern Nordic lifestyles.
Fast & Free Delivery
Get your favorite products delivered safely to your doorstep within 2-4 business days.
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Every item is rigorously tested to ensure our high Nordic standards are consistently met.
100% Secure Payments
Your transactions are protected with industry-leading encryption and top security protocols.
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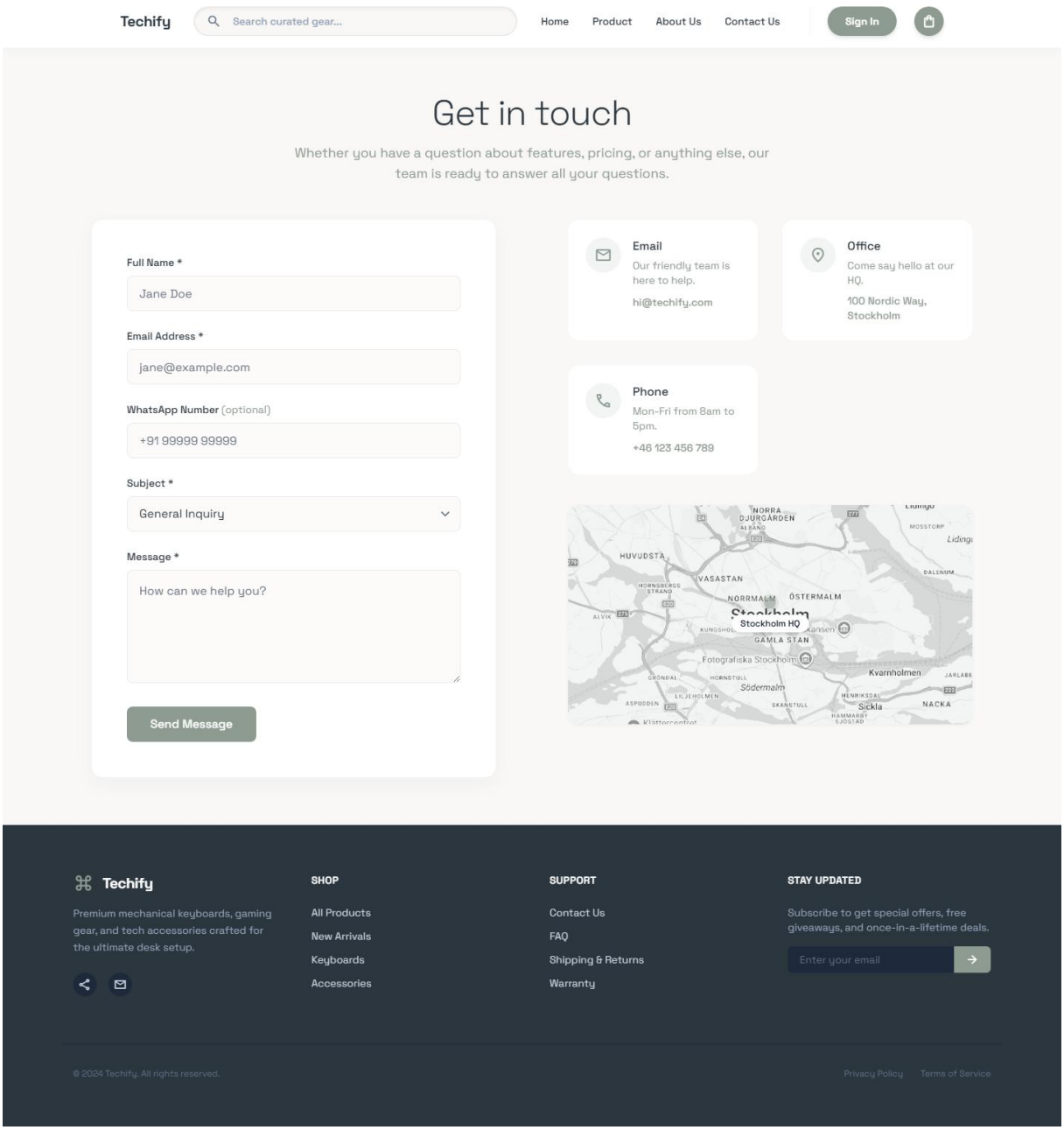
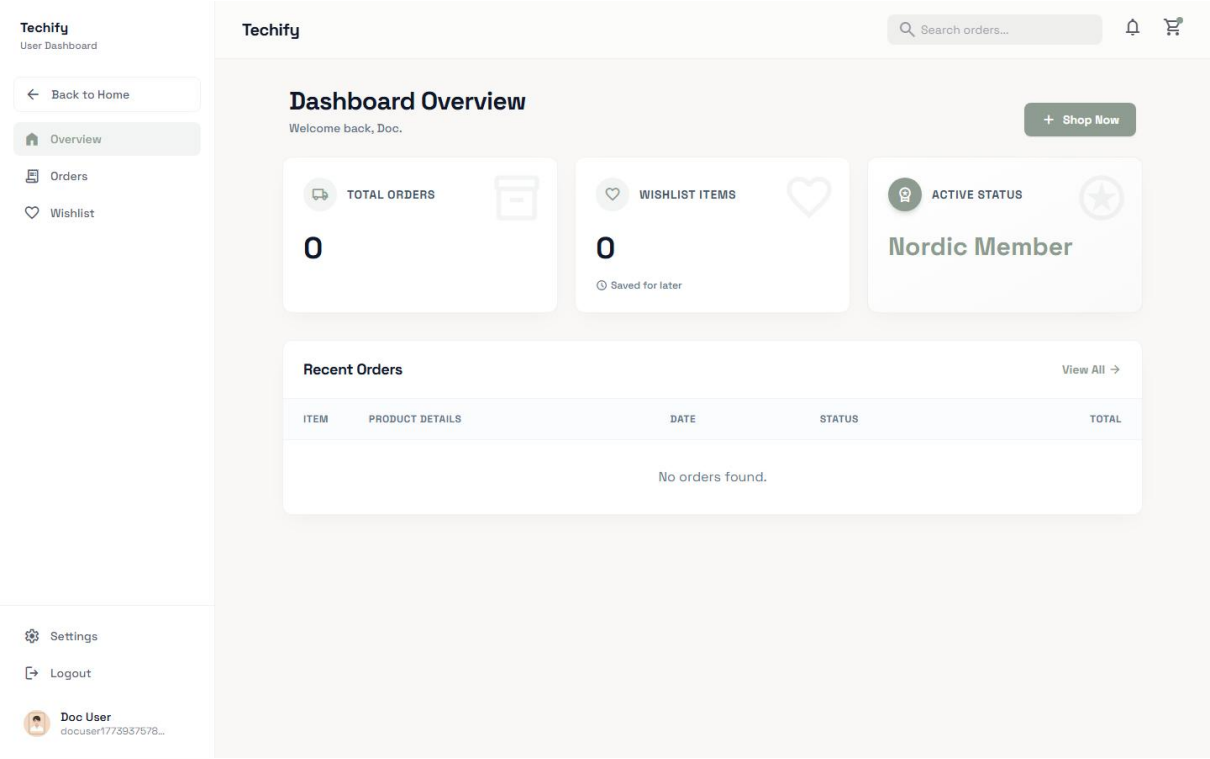
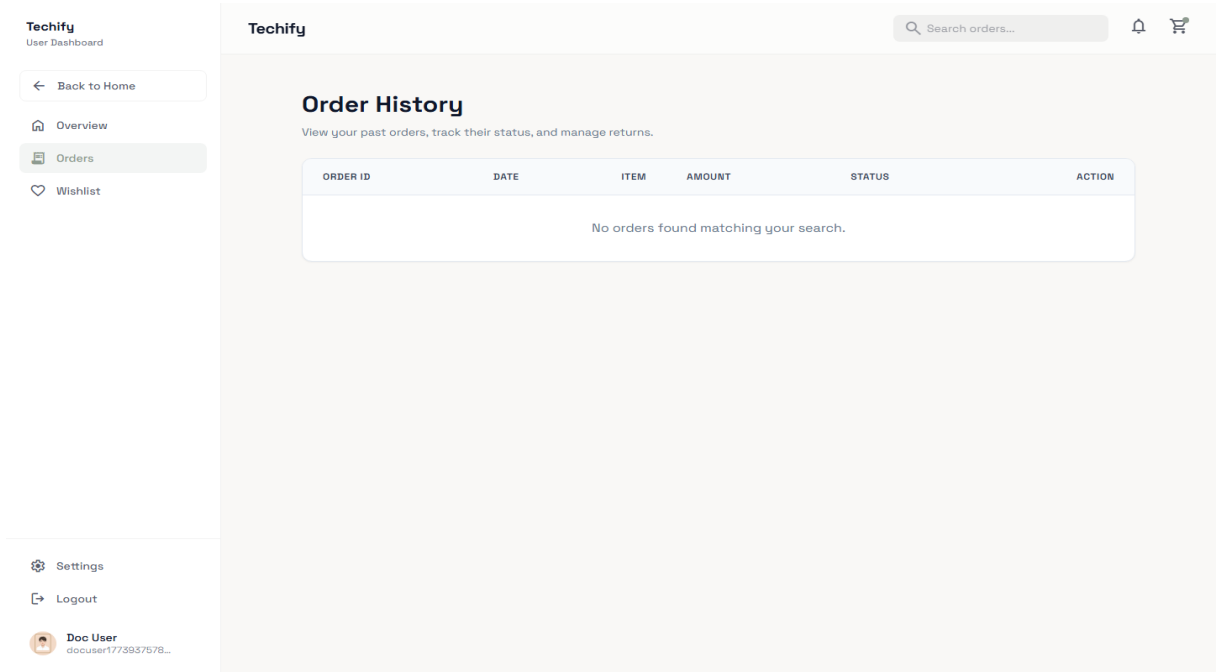
❖ Contact Page:-



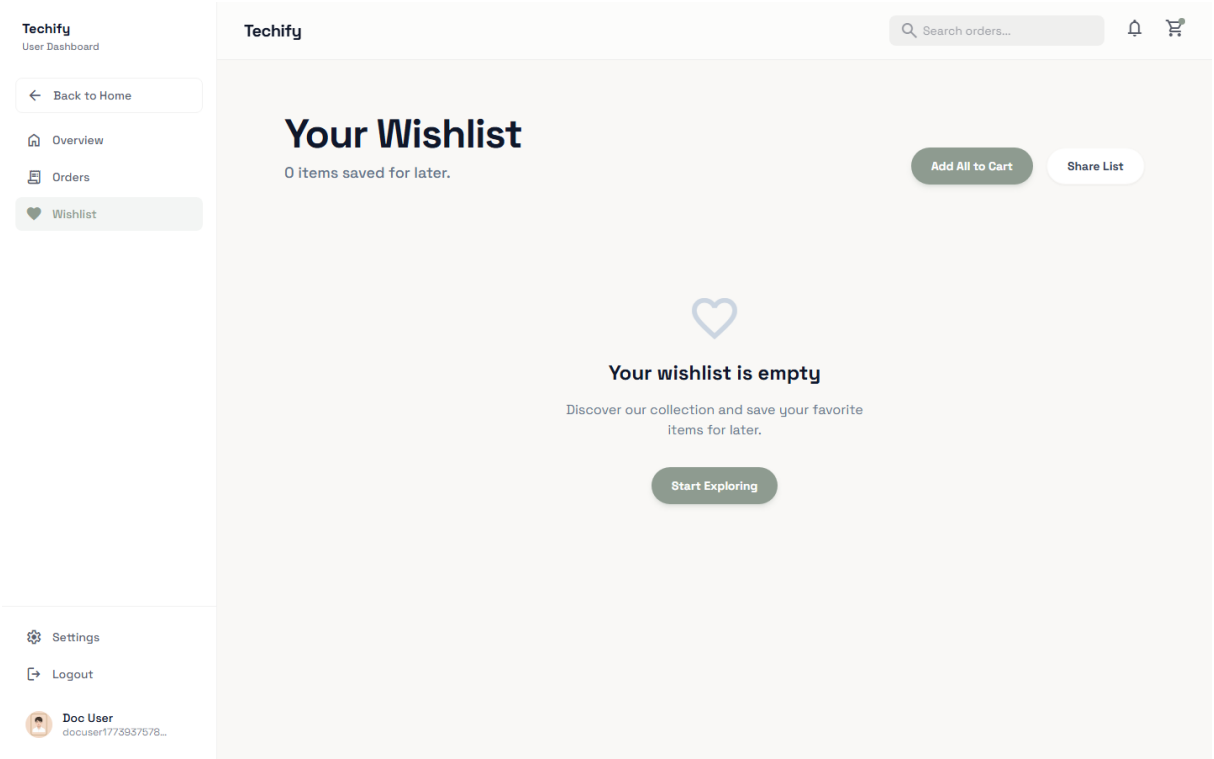
❖User Dashboard Overview :-



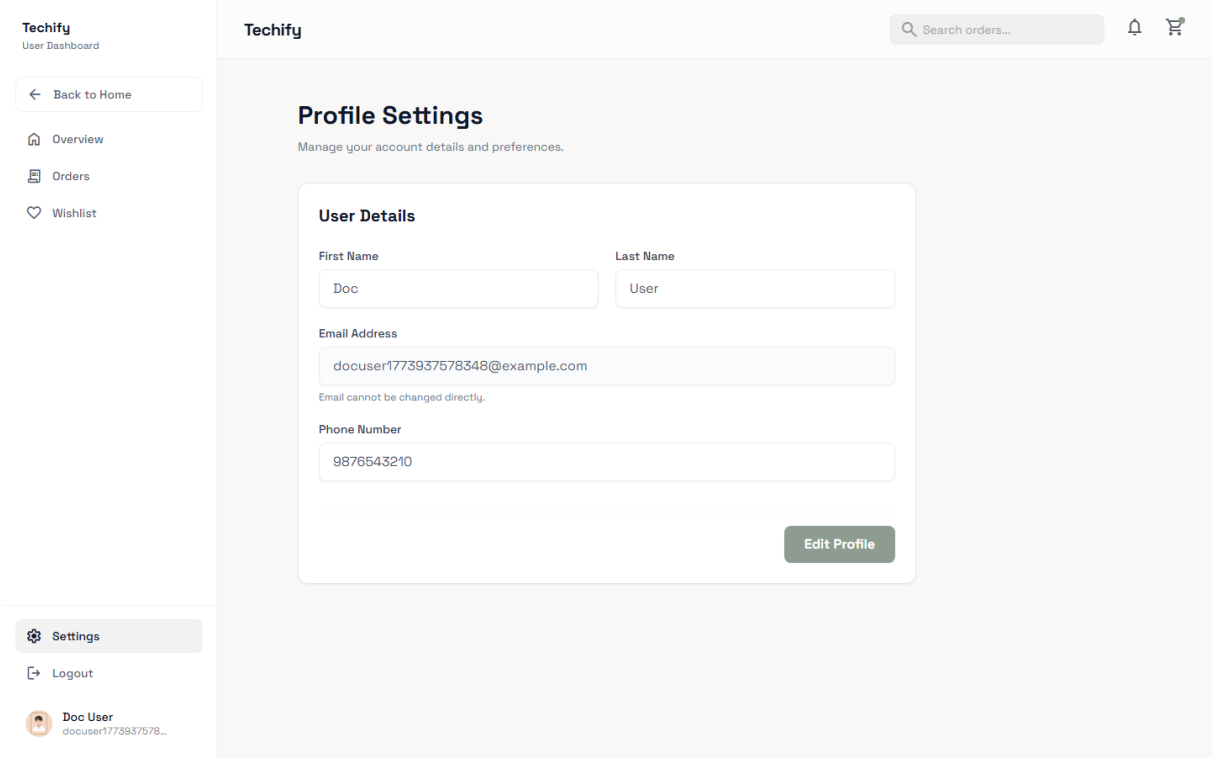
❖ User Dashboard Orders:-



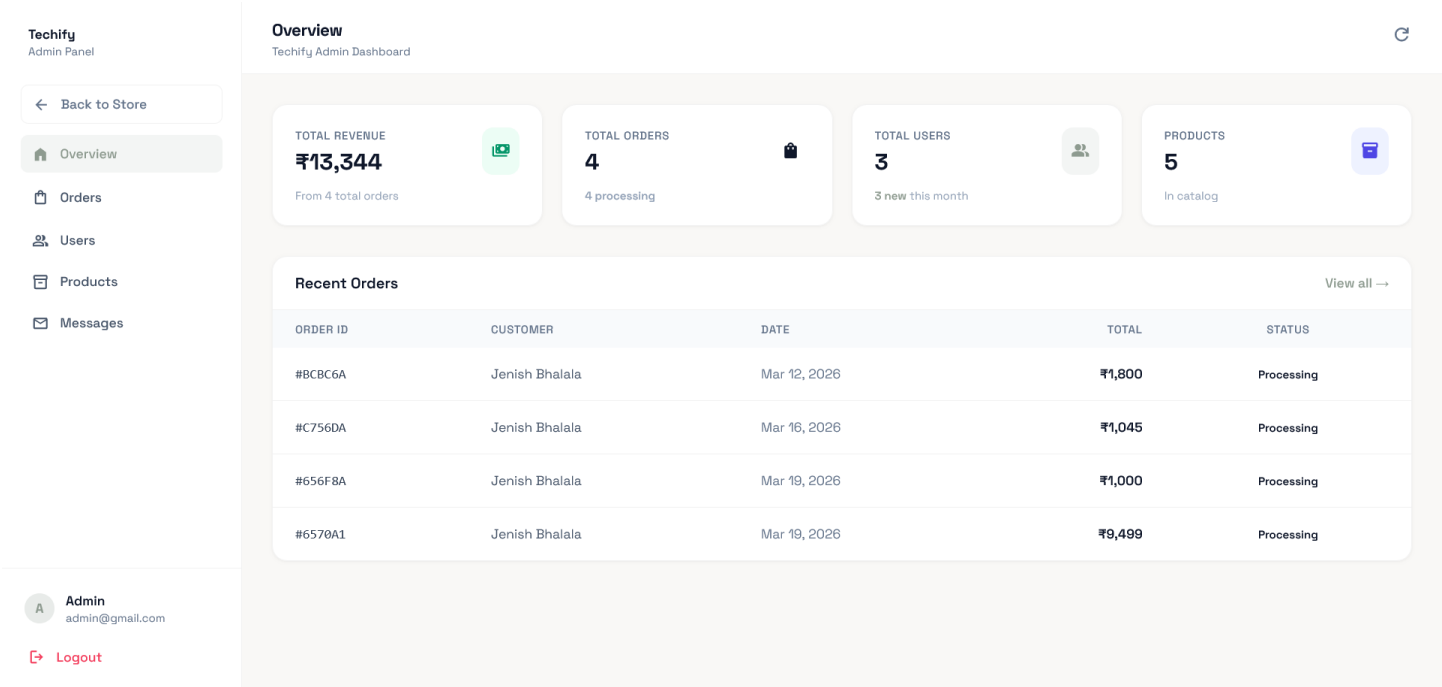
❖ User Dashboard Wishlist :-



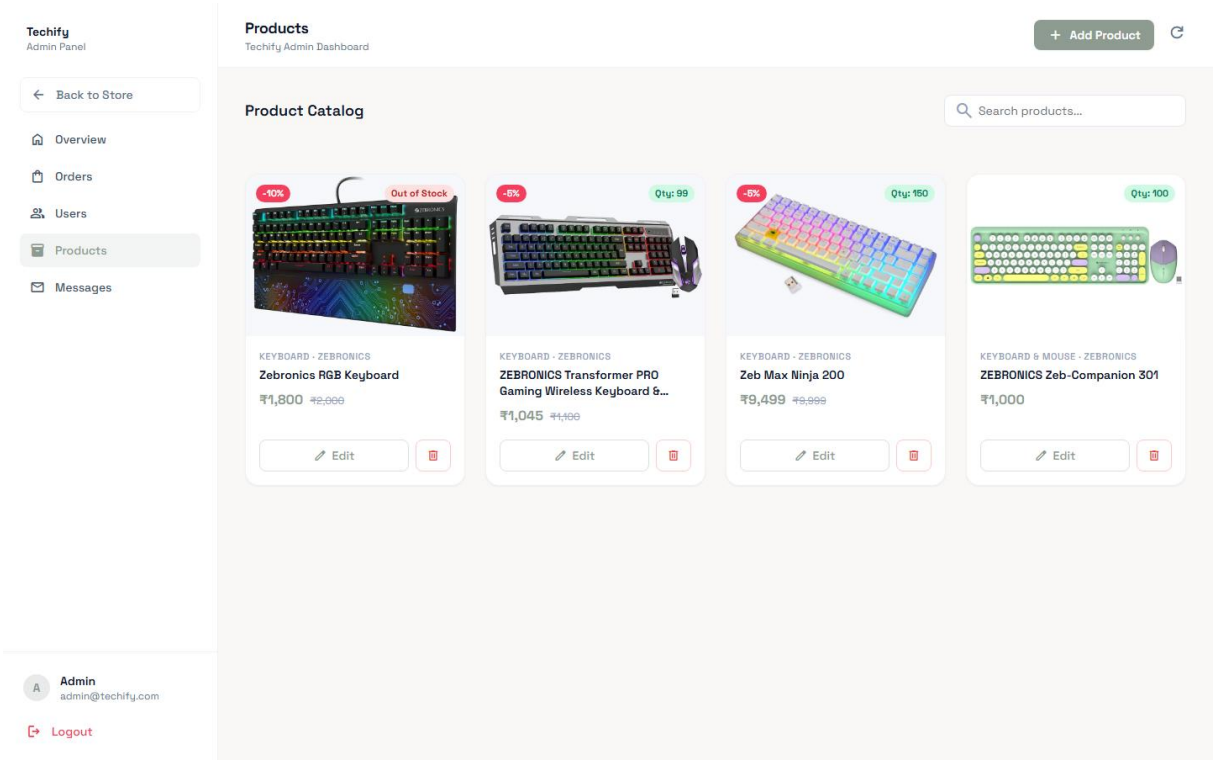
❖ User Dashboard Settings :-



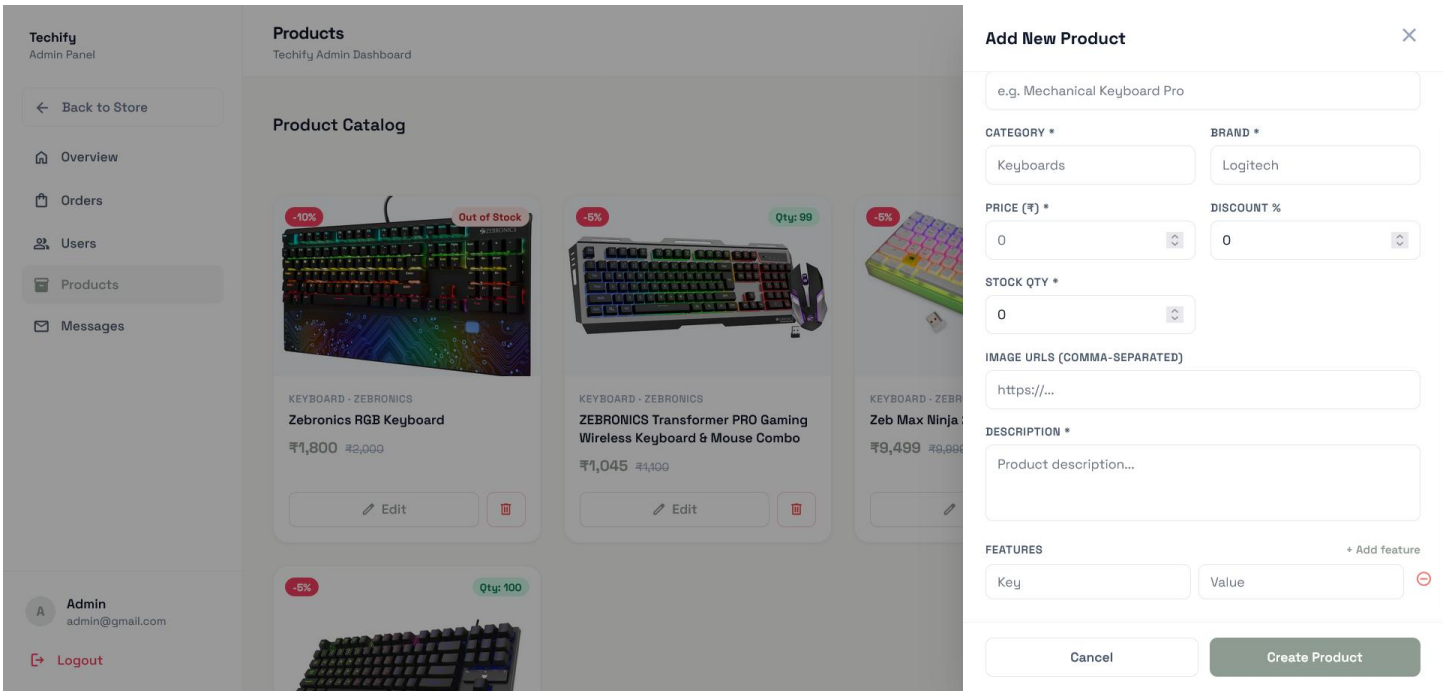
❖ Admin Dashboard Overview :-



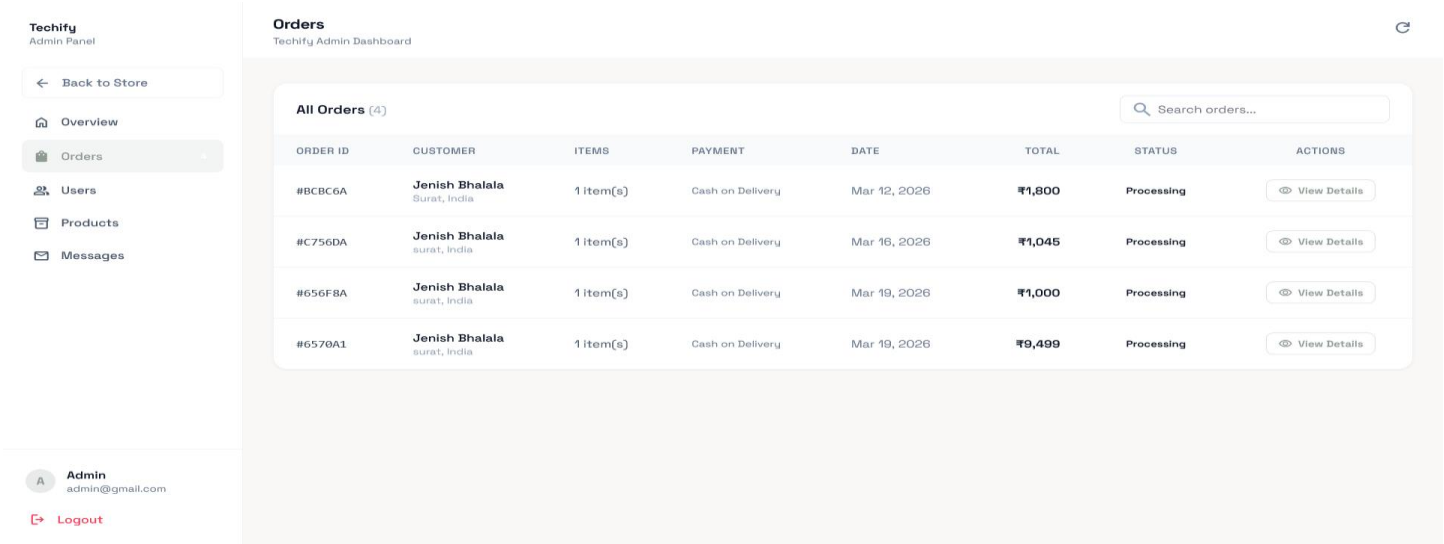
❖Admin Products Section:-



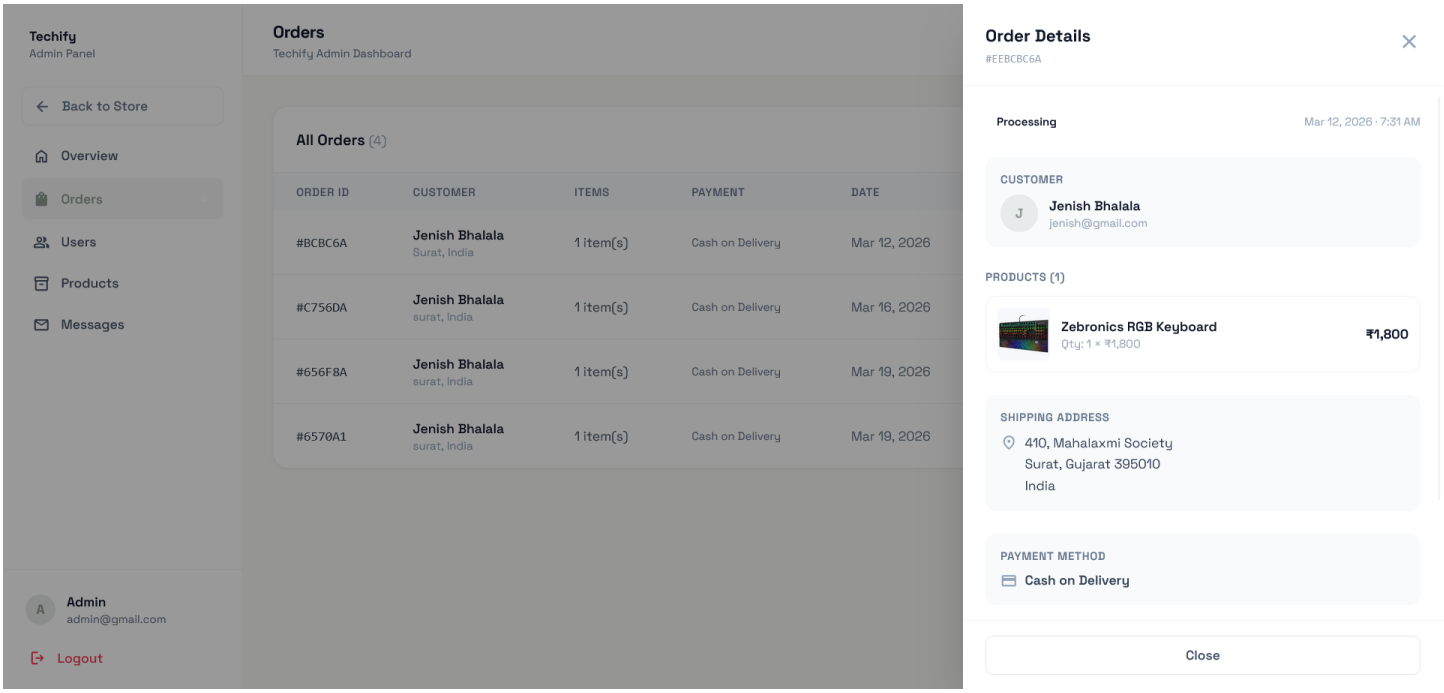
❖Admin Add Products Section:-



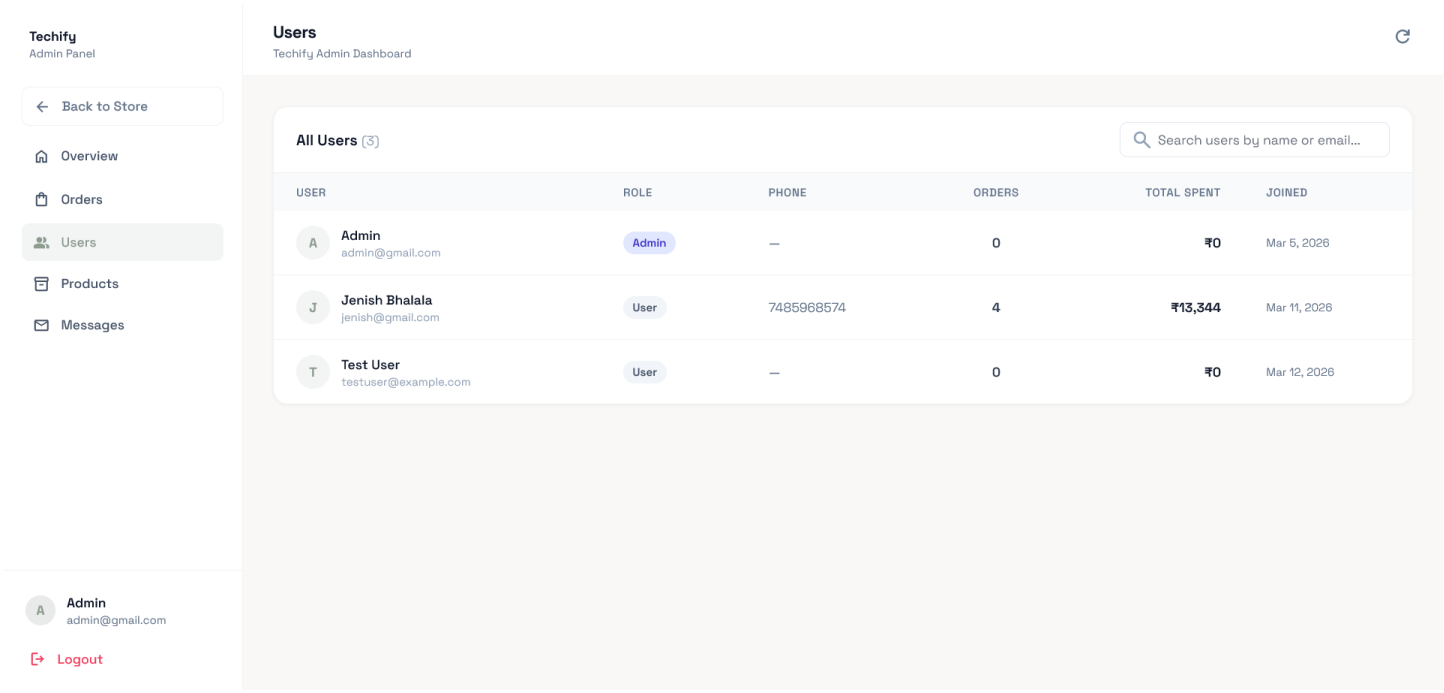
❖ Admin Orders Section:-



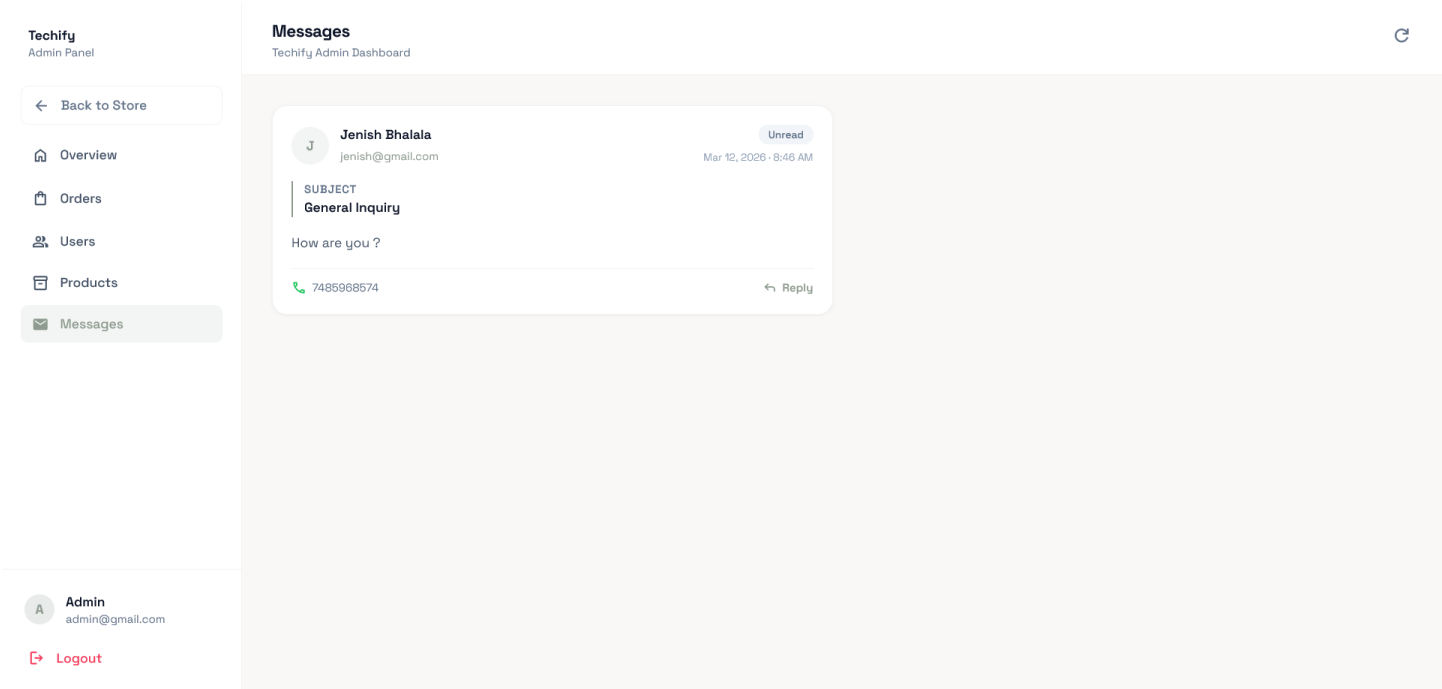
❖Admin Orders Details Section:-



❖Admin Users Section:-



❖Admin Messages Section:-



7. References

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- Node.js Documentation - https://nodejs.org/en/docs
- Express Documentation - https://expressjs.com
- MongoDB Documentation - https://www.mongodb.com/docs
- Mongoose Documentation - https://mongoosejs.com/docs
- Tailwind CSS Documentation - https://tailwindcss.com/docs
- TypeScript Documentation - https://www.typescriptlang.org/docs
- GitHub - https://github.com